

Within-Site Variation in Understory Kelp Density

Understory kelp abundance is measured along six 30×2 m transects at each site. Each transect therefore surveys an area of 60 m^2 . For species s , site i , year t , and transect j , transect-level density is defined as

$$d_{s,i,t,j} = \frac{N_{s,i,t,j}}{60}, \quad (1)$$

where $N_{s,i,t,j}$ is the number of individuals observed along transect j . The mean density across the six transects is

$$D_{s,i,t} = \frac{1}{6} \sum_{j=1}^6 d_{s,i,t,j}. \quad (2)$$

We use the following metrics to characterize variation in understory kelp abundance among transects within a site.

Standard deviation

The sample standard deviation in transect-level density is

$$SD_{s,i,t} = \sqrt{\frac{1}{5} \sum_{j=1}^6 (d_{s,i,t,j} - D_{s,i,t})^2}. \quad (3)$$

Standard deviation has the same units as density and describes the typical amount by which individual transect densities differ from the site-level mean. Larger values indicate greater spatial heterogeneity in kelp density within the site.

Variance

The sample variance in transect-level density is

$$V_{s,i,t} = \frac{1}{5} \sum_{j=1}^6 (d_{s,i,t,j} - D_{s,i,t})^2. \quad (4)$$

Variance is the squared standard deviation and provides a formal measure of dispersion among transects. Although its squared units make it less intuitive than standard deviation, it is useful in statistical analyses and for comparing changes in within-site variability through time.

Transect occupancy

Transect occupancy is defined as the proportion of transects on which the species is present:

$$O_{s,i,t} = \frac{1}{6} \sum_{j=1}^6 I(N_{s,i,t,j} > 0), \quad (5)$$

where $I(\cdot)$ equals 1 when the condition is true and 0 otherwise. Occupancy ranges from 0 to 1 and may be multiplied by 100 to express the percentage of occupied transects. This metric indicates how broadly a species is distributed across the site, independently of its mean density.

Variance-to-mean ratio

Because all transects survey the same area, the variance-to-mean ratio can be calculated from the raw transect counts:

$$VMR_{s,i,t} = \frac{\frac{1}{5} \sum_{j=1}^6 (N_{s,i,t,j} - \overline{N}_{s,i,t})^2}{\overline{N}_{s,i,t}}, \quad (6)$$

where

$$\overline{N}_{s,i,t} = \frac{1}{6} \sum_{j=1}^6 N_{s,i,t,j}.$$

The variance-to-mean ratio, also called an index of dispersion, characterizes the spatial distribution of counts among transects. Values greater than 1 indicate that counts are more aggregated or patchy than expected under a Poisson distribution, whereas values near 1 are consistent with Poisson-like dispersion. The metric is undefined when the mean count is zero and should be interpreted cautiously when mean abundance is very low.

Together, these metrics distinguish changes in average kelp density from changes in its spatial distribution. Standard deviation and variance quantify the magnitude of variation among transects, occupancy measures the spatial extent of occurrence within the site, and the variance-to-mean ratio describes the relative aggregation of individuals among transects.